

VENKATA KESAV VENNA

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Education

International Institute of Information Technology, Hyderabad

2021 – 2025

B.Tech in Computer Science & Engineering

8.88/10 CGPA

Work Experience

BharatGen

May 2024 – Present

Research Engineer

- Core contributor for *Patram-Core*, a modular VLM training framework scaling to **512 GPUs** with plug-and-play vision encoders, projectors, and LLMs; used to train *Patram-7B-Instruct* via a custom curriculum on Indian documents spanning **23 languages**. [\[Link\]](#)
- Led research on grounded document understanding (*M3Grounder*, *REPLICA*, *DoCoG*, *DocLayout-VL*): 1 accepted at CVPR 2026, 2 accepted at ICDAR 2026, 2 under review at ECCV.

Centre for Visual Information Technology, IIIT-H

March 2023 – March 2024

Research Intern · Advised by Dr. Ravi Kiran (IIIT-H) & Dr. Ganesh Ramakrishnan (IIT Bombay)

- Built document understanding pipelines for digitizing India's circular library; sponsored by the Government of India.

Publications

M3Grounder: Mask-Based Multi-Span and Multi-Granular Grounding for Document QA

CVPR 2026 (**CORE: A***)

1st Author

- Novel VLM-segmentation model grounding answer spans at phrase, line, and block granularities via pixel-level masks; state-of-the-art on **4 grounding benchmarks**, improving $F1_g$ from 66.4 to **73.3** on DOGR-Bench at 8B scale. [\[Link\]](#)
- Released *GroundingDocQA: 200K documents, 2M QA pairs* with pixel-level masks, and *GroundingDocQA-Bench*, a 2.5K-document human-verified benchmark covering curved text and multi-span settings.

DocLayout-VL: Foundational Model for Hierarchical, Open-Set & Promptable Layout Segmentation

ECCV 2026 (under review) (**CORE: A***)

1st Author

- Foundation model for document layout understanding via promptable, open-set, hierarchical segmentation; achieves **79.4 OS** vs. 37.2 for Gemini 3 Pro. Novel HGC enforces parent-child-sibling consistency via tree-aware sparse attention.

REPLICA: An Agentic Framework for Visually Faithful Document Reconstruction

ICDAR 2026 (**CORE: A**)

1st Author

- Agentic PDF-to-HTML framework (Segment → Localize → Assemble → Refine) achieving **93% visual fidelity** across **15+ languages** and **20+ document types**, outperforming GPT-5 and Gemini 3 Pro. REPLICA with Qwen3-VL-8B achieves **39% gain** over Qwen3-VL-30B despite being $3.75\times$ smaller. Introduced *Fid-HTML* and *VFDR-Bench* (5K expert-annotated documents).

DoCoG: Mask-Based Multi-Type Grounded Chain-of-Thought for Document QA

ECCV 2026 (under review) (**CORE: A***)

- Step-wise grounded CoT architecture with a novel Grounding Interaction Module (GIM) for cross-[GND]-token consistency; DPO-based alignment with anchor loss preserves mask fidelity during preference optimization. Achieves **85.8 AQ** and **81.9 GQfa** vs. 84.2 AQ and 11.4 GQfa for Gemini 3 Pro—first use of DPO in grounded DocVQA.
- Introduced *DoCoG-QA (325K documents, 1.5M step-wise grounded QA pairs* across charts, tables, flowcharts) and *DoCoG-PQA* (20K preference pairs for alignment).

Patram-Bench: Multi-task, Multi-domain, Multi-lingual Benchmark for Indian Document Understanding

ICDAR 2026 (**CORE: A**)

- First large-scale multilingual Indic doc benchmark: **52K documents, 23 languages, 13 domains, 71 classes, 52 tasks, 2.4M QA annotations** across OCR, layout, structured parsing, multi-hop VQA. Introduced DUCLI and SRS metrics; evaluated **29 VLMs**.

MAdVerse: A Hierarchical Dataset of Multi-Lingual Ads from Diverse Sources and Categories

WACV 2024 (**CORE: A**)

- Fine-tuned CNN and transformer classifiers for hierarchical multilingual ad classification; built a newspaper ad extraction pipeline using Faster R-CNN and ViT on a large multilingual corpus. [\[Link\]](#)

Projects

GPU Programming — Ongoing Kernel Series

2026 – Present

CUDA C/C++ · CSE 234 · PMPP · GPU Mode · Ongoing

- Built a progression of 9 CUDA kernels from scratch: vector addition → matrix addition → matrix-vector multiply → naive SGEMM → coalesced SGEMM → tiled shared-memory SGEMM → 1D register-tiled SGEMM; each step benchmarked and analyzed for warp-level memory access patterns and arithmetic intensity. [\[Github\]](#)
- Implemented naive $O(N^2)$ attention and Flash Attention forward pass from scratch in CUDA; fused score, online softmax, and output accumulation into a single kernel pass, reducing HBM memory complexity from $O(N^2)$ to $O(N)$; validated both against PyTorch.